

Course syllabus for

Academic year 2022/2023 >

MCCX11 - Bachelor's thesis in Microtechnology and nanoscience

Kandidatarbete vid Mikroteknologi och nanovetenskap

Course syllabus adopted 2022-02-07 by Vice President (or corresponding)

Owner: [TKELT](#)

15,0 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: First-cycle

Main field of study: Automation and Mechatronics Engineering, Electrical Engineering, Chemical Engineering with Engineering Physics, Mathematics, Biomedical engineering, Engineering Physics

Department: 59 - MICROTECHNOLOGY AND NANOSCIENCE

Teaching language: Swedish

Application code: 50143

Open for exchange students: No

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course	No Sp	Examination dates
0122 Project 15,0 c Grading: TH			7,5 c	7,5 c			

In programs

[TKAUT AUTOMATION AND MECHATRONICS ENGINEERING, Year 3](#)

[TKELT ELECTRICAL ENGINEERING, Year 3](#)

[TKKEF CHEMICAL ENGINEERING WITH ENGINEERING PHYSICS, Year 3](#)

[TKMED BIOMEDICAL ENGINEERING, Year 3](#)

[TKTFY ENGINEERING PHYSICS, Year 3](#)

Examiner:

[Per Lundgren](#)



[Go to course homepage](#)

Eligibility

General entry requirements for bachelor's level (first cycle)

Specific entry requirements

The specific entry requirements are the same as for the programme the course is taken within

Course specific prerequisites

The bachelor thesis work can commence no earlier than during the 6th programme semester. The programme director decides which bachelor thesis projects that are allowed for the students on the respective programme. In order to secure a sufficient level of prior knowledge, students must have attained at least 105 c from courses in the first three years of their programmes to commence their bachelor's thesis work. Only credits registered in LADOK after the exam period 1 in study year 3 will be taken into consideration, however, at the latest when the project selection closes.

Aim

In the bachelor thesis project, the student will integrate, extend and develop knowledge and skills within a smaller, specialized part of the field of the programme. The bachelor thesis also aims to develop skills in application of engineering problem solving and research methods.

Learning outcomes (after completion of the course the student should be able to)

Knowledge, understanding, competence and skills

- Formulate and delimit a problem within the chosen subject
- Plan a project in order to solve and report the problem with given resources
- Search for, compile and assess relevant literature and other background information
- Integrate and develop knowledge relevant for the chosen project
- Reflect on how the project group worked together to meet common goals
- Document the work process in a project journal

- Communicate the results of the bachelor thesis work in relation to the chosen problem in written and oral form

Judgement and approach

- Critically examine, evaluate and constructively question another project with respect to problem, approach and results
- Assess if societal and ethical issues need to be considered for the chosen problem and, where relevant, analyze these aspects in the essay/report
- Reflect on the need for further knowledge and provide suggestions for future projects in the subject

Content

The main part of the bachelor thesis is a project. The student chooses the specialization topic by selecting a project approved for the programme. To facilitate and support the project, information and communication skills are also included. Particular emphasis is given to practice to report one's work, both orally and by independently writing a part of an essay/report that shall be worth reading for an imaginary relevant readership.

Organisation

The bachelor thesis is conducted mainly as a project and the work is conducted in a group of up to six people. At given times tutorials, within the projects' specialization topic and in information and communication skills, are provided. During the project, progress reports are to be handed in for continuous control and feedback.

Examination including compulsory elements

The Bachelor's thesis is presented in a written essay/report in accordance with current standards of scientific and technical reports and orally presented and defended at a seminar. Each student shall write a clearly identifiable part of a joint essay/report which should be described in the project journal. In addition, the student should also critically examine and discuss another group's thesis project in a written opposition and by serving as an opponent at the other group's presentation seminar. For the examination, emphasis is placed on the quality of the project report, project journal, oral presentation and opposition, especially linked to the general objectives of the bachelor thesis. For approved bachelor's thesis, a pass is also required in the sections on information and communication skills.

The examiner is responsible for the report to be checked in a plagiarism tool

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.